Dual Nature of Radiation & Matter

⇒ Newton (1670)
- Light is a particle (Corpuscular particle).
- Explain reflection / Refraction.

⇒ Huygen (1690)
- Light is a mechanical wave.
- Explain Interference / diffraction / Polarisation
- Ether medium for light waves.

⇒ Maxwell (1865)
- Light is a non-mechanical waves
- No medium require
- Transverse wave in nature
- E.M wave explain.

⇒ Max. Plank (1901)
- Light is a particle, $E = hf$
- Photon
- Black body radiation / Photoelectric effect / compton effect.

⇒ De-Broglie (1923)
- Light have dual Nature → (Behave as ray as well as wave or particle.

⇒ Davission & Germer Expt (1927)
- e^- is a wave

⇒ G.P thomoson
- Get nobel prize for experiment verification of e^- as a wave.

Particle Nature of Light :-

↳ Collection of tiny particle called photon.

$$\boxed{E = h\nu = \frac{hc}{\lambda} = \frac{2\times10^{-25}\,J}{\lambda(m)} = \frac{12400\,eV}{\lambda(A^0)}}$$

h = plank's const. $= 6.63\times10^{-34}$ J-S

$$\boxed{1eV = 1.6\times10^{-19}\,J}$$

Energy of Light = Energy of Radiation

$$E = \frac{nh\nu}{\lambda} = \frac{nhc}{\lambda}$$ n = no. of photons

Properties of Photon

1. Charge of Photon = 0 (neutral)
2. Rest Mass of photon = 0
3. Speed of photon in vaccume

$$C = \frac{1}{\sqrt{\mu_0\varepsilon_0}} = 3\times10^8\ m/s$$

4. Speed of photon in medium

$$V = \frac{C}{\mu} = \frac{C}{\sqrt{\mu_r\varepsilon_r}} = \frac{1}{\sqrt{\mu_m\varepsilon_m}}$$

5. Photon can't be charged
6. It can't deviates in E.M field.
7. X-ray is a collection of photon.

Power of light (for Beam light)

$$\boxed{P = \frac{E}{t} = \frac{nh\nu}{t} = \frac{nhc}{t\lambda}}$$

no. of photon per sec

$$\boxed{\frac{n}{t} = \left[\frac{P\lambda}{hc}\right]}$$

Momentum of photon

$$E = PC$$

$$\boxed{P = mc = \frac{E}{C} = \frac{h\nu}{c} = \frac{h}{\lambda}}$$

~~# Total~~ # Intensity of light

↳ Total energy of light falling per-sec per unit area.

$$\boxed{I = \frac{nE}{At} = \frac{nh\nu}{At} = \frac{nhc}{\lambda At} = \frac{Power}{Area}}$$

Note:-
- Frequency does not depends upon medium & distance from source, only depend upon nature of source.
- Nothing is given about source, source remains same (ν = same)

$I \propto$ no. of photon, $I \propto n$

- source is variable (ν = changing)

$I \propto n\nu$

Radiation pressure

- Complete Absorptive :- ($\rho = 1$)

$$P = \frac{I}{c}[2-\rho]\ , \quad P = \frac{I}{c}$$

$$\boxed{\text{Force} = PA = \frac{IA}{c} = \frac{\text{Power}}{c}}$$ P = Pressure

★ Complete reflective :- ($\sigma = 1$)

$P = \frac{I}{c}[\sigma + 1]$, $P = \frac{2I}{c}$

$$\boxed{\text{force} = \frac{2IA}{c} = \frac{2\text{Power}}{c}}$$

σ = reflection of coefficient

$\sigma = 0$ (No reflection)

$\sigma = 1$ (complete reflection)

$\sigma = \frac{1}{2}$ = 50% reflection

Q ⇉ If Intensity of light is I & 75% radiation is absorbed then find Radiation pressure?

Sol^n - 75% absorbed means 25% reflected

therefore, $P = \frac{I}{c}(\sigma + 1) = \frac{I}{c}\left(\frac{1}{4} + 1\right)$

$P = \frac{5I}{4c}$ Ans

Photoelectric Effect

- Ejection of e^- from metal surface, by the falling of light of sufficient energy.

Photon (light energy) ⇉ Electron [Electric Energy]

★ law of photoelectric effect

- Instaneous process, $t = 10^{-9}$ sec (b/w falling of photon & ejection of e^-)
- Efficiency of photoelectric $= \frac{\text{no. of electron}}{\text{no. of photon}}$
- one to one interaction $= 10^{-3}$ to 10^{-4}.

one photon → one electron.

- Let 1e⁻ is ejected → $\frac{1}{\text{no. of photon}} = 10^{-3}$

$\therefore$ no. of photon $= 10^3$.

- $\boxed{K.E_{max} = E - \phi}$
- stopping potential, $\boxed{eV_0 = K.E_{max}}$

★ work function (ϕ) :-

↳ Min. energy required to pull the e^- from material.

- ~~These energy required to pull the electron~~
- These energy can be given by:-

① Thermionic emission (By heat)

② field emission (By electric field)

③ Secondary emission (By collision of e)

④ Photoelectric effect (By light)

$$\boxed{\phi = h\nu_0 = \frac{hc}{\lambda_0}}$$ → only depends on nature of Metal. → Unit → eV.

ν_0 → thereshold frequency

λ_0 → thereshold wavelength

$$\boxed{E_{photon} = \phi + (K.E)_{max}}$$ → Einstein equ^n of photoelectric effect

- $h\nu = h\nu_0 + eV_0$
- $\frac{hc}{\lambda} = \frac{hc}{\lambda_0} + \frac{1}{2}m_e v_{max}^2$

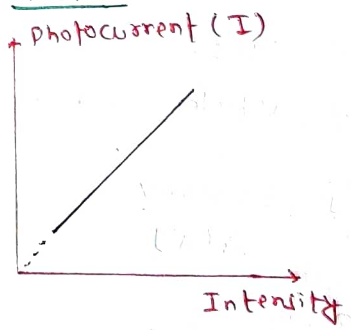
★ for PEE

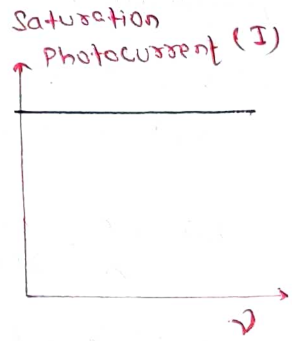
$$\boxed{\nu_{light} \geq \nu_0 \quad , \quad \lambda_{light} \leq \lambda_0}$$

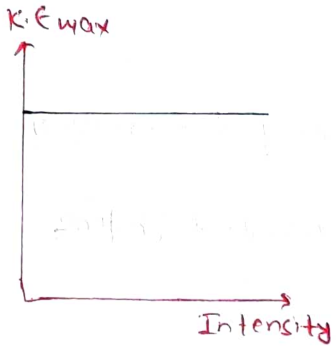
NOTE :-

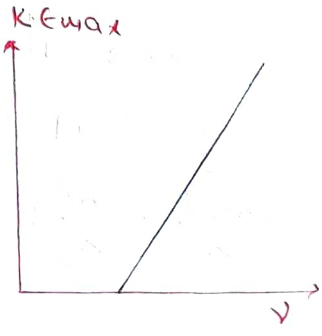
- Intensity ∝ no. of photon ∝ no. of electron ∝ photocurrent
- frequency ∝ energy of photon ∝ energy of e^- ∝ (K.E) ∝ stopping potential
- Photocurrent depends on intensity not on frequency.
- Stopping potential depends on frequency not on intensity.

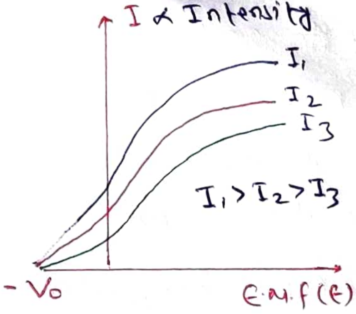
★ GRAPh :-

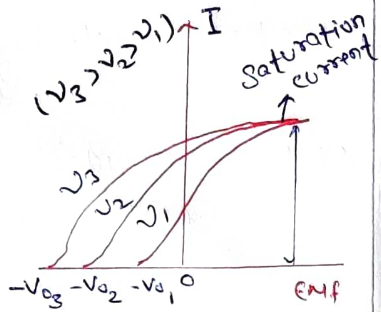






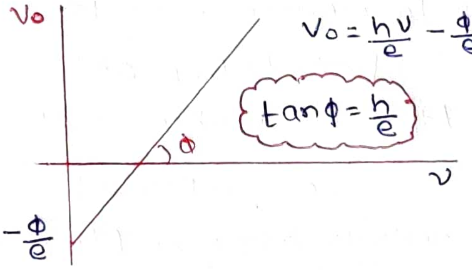






$V_0 = \frac{h\nu}{e} - \frac{\phi}{e}$

$\tan\phi = \frac{h}{e}$



Matter Wave

↳ Every matter have dual nature.

★ De-broglie wavelength of Matter wave

$$\boxed{\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2m(K.E)}} = \frac{h}{\sqrt{2mqV}}}$$ → Valid for e^-, p^+, not for photon.

for proton, $\lambda_p = \frac{0.286}{\sqrt{V}}$ A°

for Electron, $\lambda_e = \frac{12.27}{\sqrt{V}}$ A°

for deutron, $\lambda_d = \frac{0.202}{\sqrt{V}}$ A°

for neutron, $\lambda_n = \frac{0.286}{\sqrt{E}}$ A° (E in eV)

for α-particle, $\lambda_\alpha = \frac{0.101}{\sqrt{V}}$ A°

for gas molecule, $\lambda_{gas} = \frac{h}{\sqrt{3mk_BT}}$

$\lambda = \frac{h}{\sqrt{2m\left(\frac{13.6}{n^2}\right)}}$

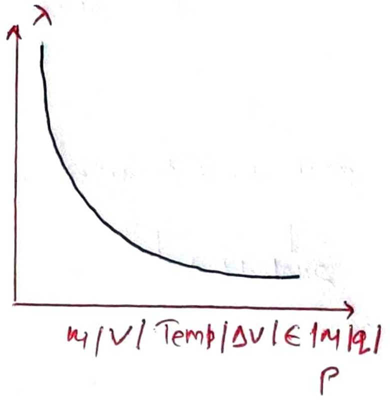
$\lambda = \frac{h}{p} = \frac{h}{qBR}$

$\lambda = \frac{h}{mv}$

$\lambda = \frac{h}{p} = \frac{h}{Ft} = \frac{h}{qEt}$

$\lambda = \frac{h\sqrt{1 - v^2/c^2}}{m_0 v}$

$\lambda = \frac{h}{\sqrt{2mq\Delta V}}$



★ Heisenberg principle

$\Delta x \cdot \Delta P \geq \frac{h}{4\pi}$

$\boxed{\frac{h}{2\pi} = \hbar}$

$\Delta L \cdot \Delta\theta \geq \frac{h}{4\pi}$